

Advanced (Habanero)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-1/2$
(B) $1/2$
(C) 2
(D) -2
(E) 1
- Q2.** Which of the following lines is perpendicular to $y = -3x - 2$?
(A) $y = 1/3x + 1$
(B) $y = -1/3x - 1$
(C) $y = 3x - 1$
(D) $y = -3x + 1$
(E) $y = 1/3x - 1$
- Q3.** The equation of a line perpendicular to $y = x - 2$ is
(A) $y = -x - 2$
(B) $y = -x + 2$
(C) $y = x + 2$
(D) $y = -x + 1$
(E) $y = x - 1$
- Q4.** If $y = 2x + 1$ and $y = mx - 1$ are perpendicular, what is m ?
(A) $-1/2$
(B) $1/2$
(C) -2
(D) 2
(E) 0
- Q5.** The slopes of two perpendicular lines are m_1 and m_2 . What is $m_1 \cdot m_2$?
(A) 1
(B) -1
(C) 0
(D) 2
(E) -2
- Q6.** What is the equation of a line perpendicular to $y = -2x + 1$ and passing through $(1, 2)$?
(A) $y = 1/2x + 3/2$
(B) $y = -1/2x + 5/2$
(C) $y = 2x - 1$
(D) $y = -2x + 3$
(E) $y = 1/2x + 1/2$
- Q7.** If two lines have slopes 2 and $-1/2$, they are
(A) parallel
(B) perpendicular
(C) neither parallel nor perpendicular
(D) the same line
(E) skew lines
- Q8.** The equation of a line passing through $(2, 1)$ and perpendicular to $y = 3x - 2$ is
(A) $y = -1/3x + 5/3$
(B) $y = -1/3x + 7/3$
(C) $y = 3x - 1$
(D) $y = 3x + 1$
(E) $y = -1/3x - 1/3$

Open Ended Questions (FRQ)

- Q9.** Find the equation of a line passing through $(3, -2)$ and perpendicular to $y = 2x + 1$. Show all work.
- Q10.** Prove that the lines $y = x + 2$ and $y = -x - 1$ are perpendicular. Use the definition of perpendicular lines to show your work.

Chef's Tips

- Provide a calculator for students
- Make sure students have extra paper for work
- Emphasize the importance of showing all work for free response questions